



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

REVIEW TEST

19-12-2021

JEE(Main + Advanced) : NURTURE COURSE [PHASE : S(TEPS)]

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	C	A	C		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	B,C	A,C,D	A,B,C,D	A,D	A,B,C	C
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	10.00	4.00	0.20	4000.00	370.00	1075.00
SECTION-II (ii)	Q.	7	8	9			
	A.	6	3	3			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	A	D	C		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,D	A,C	A,B,D	B,C	A,C	A,B,C,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	0.66 to 0.67	2.00	3.00	5.00	48.00	54.00
SECTION-II (ii)	Q.	7	8	9			
	A.	6	9	3			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	A	B	A	D		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	B	A,B,C,D	A,B,C	C,D	A,B,C	A,B,C,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	10.00	1.20	32.00	8.00	2.00	0.00
SECTION-II (ii)	Q.	7	8	9			
	A.	7	1	3			

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (B)

$$C = C_V + \frac{R}{1-n} \text{ in the given process}$$

$$P \propto V$$

$$PV^{-1} = \text{const}$$

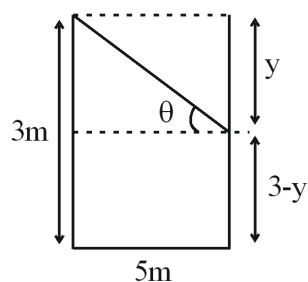
$$n = -1$$

$$\Rightarrow C = \frac{3}{2}R + \frac{R}{1-(-1)} = 2R$$

2. Ans (C)

x, r & d all increase

3. Ans (A)



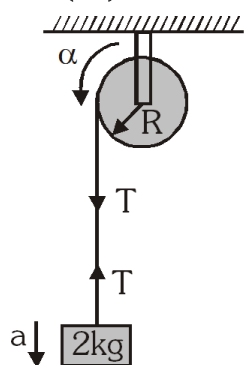
$$\left((3-y)5 + \frac{1}{2} \times y \times 5 \right) 4 = 5 \times 2 \times 4$$

$$3 - y + \frac{y}{2} = 2 \frac{y}{2} = 1$$

$$y = 2 \frac{2}{5} = \frac{4}{5}$$

$$a = 4\text{m/s}^2$$

4. **Ans (C)**



For motion of block $2g - T = 2a$
For motion of pulley $\tau = TR = Ia$

$$\therefore a = \alpha R \therefore T = \frac{Ia}{R^2} \Rightarrow 2g - \frac{Ia}{R^2} = 2a$$

$$\Rightarrow a = \frac{g}{1 + \frac{I}{2R^2}}$$

$$\Rightarrow a = \frac{10}{1 + \frac{0.32}{2 \times 0.2 \times 0.2}} = \frac{10}{1 + 4} \frac{10}{5} = 2 \text{ ms}^{-2}$$

PART-1 : PHYSICS

SECTION-I (ii)

6. **Ans (A,C,D)**

At near absolute zero, molecules are at lowest potential energy level. Thus from graph, A expands, B neither expands nor contract while C contracts on heating.

7. **Ans (A,B,C,D)**

$\Delta u = nC_v \Delta T$ for all processes

for adiabatic process $\Rightarrow \Delta Q = 0$

$$\Rightarrow \Delta u = \Delta \omega = 0 \Rightarrow \Delta u = -\Delta \omega$$

Here $\Delta \omega$ is work done on the gas

$\therefore -\Delta \omega = \text{work done by the gas} = \text{change in I.E.}$

8. **Ans (A,D)**

Friction will act forward. Magnitude of angular momentum about A and B are same but direction is opposite.

9. **Ans (A,B,C)**

$$T = mg, T = m\omega^2 r$$

$$K = \frac{1}{2} m(\omega \times r)^2$$

Torque of tension is always zero about hole.

10. **Ans (C)**

$$a = \frac{g \sin \theta}{1 + \frac{I}{mR^2}} = \frac{5g \sin \theta}{7}$$

$$\alpha = \frac{a}{R}$$

$$\mu_{\min} = \frac{\tan \theta}{1 + \frac{mR^2}{I}} = \frac{2 \tan \theta}{7}$$

Friction always acts in upward direction in such cases of pure rolling on inclined plane.

PART-1 : PHYSICS

SECTION-II (i)

1. **Ans (10.00)**

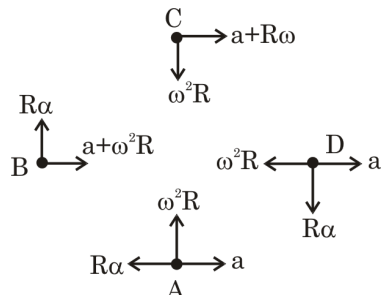
$$\vec{a}_{B/D} = 2\omega^2 R \hat{i} + 2R\alpha \hat{j} = 2(2)^2 \times 1 \hat{i} + 2 \times 1 \times 3 \hat{j}$$

$$= 8 \hat{i} + 6 \hat{j} \text{ m/s}^2$$

$$a_{B/D} = \sqrt{8^2 + 6^2}$$

$$= 10 \text{ m/s}^2$$

2. **Ans (4.00)**



$$a_A = (2)^2 \times 1 = 2 \text{ m/s}^2$$

3. **Ans (0.20)**

$$\Delta Q = ms\Delta T \Rightarrow \frac{\Delta Q}{\Delta T} = m \times 0.05$$

4. **Ans (4000.00)**

$$\Delta Q = m \times L_f$$

5. **Ans (370.00)**

$$\vec{v} = v_x \hat{i} + v_y \hat{j}; v_x = u_x \& v_y = u_y - gt$$

6. **Ans (1075.00)**

$$y = u_y t - \frac{1}{2} gt^2 = 400 \times \frac{3}{5} - \frac{1}{2} \times 10 \times 25 = 1075\text{m}$$

PART-1 : PHYSICS

SECTION-II (ii)

7. **Ans (6)**

On applying momentum conservation, common velocity after collision $\rightarrow 2\text{m/s}$

By W.E.T.

$$-(\mu mg)x - \frac{1}{2}kx^2 = \frac{1}{2}(5+1)2^2$$

8. **Ans (3)**

Slope of temperature graph = temperature gradient

$$\text{Temperature gradient} = -\frac{i}{kA} = \tan \theta$$

$$\frac{\tan \theta_1}{\tan \theta_2} = \frac{k_2}{k_1} = \frac{\sqrt{3}}{\frac{1}{\sqrt{3}}} = 3$$

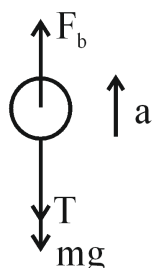
9. **Ans (3)**

$$F_b - T - mg = ma$$

$$F_b - mg - ma = T$$

$$4 \times 10^{-3} \times 10^3 \times (10+2) - 2 \times 10 - 2 \times 2 = T$$

$$T = 24$$



$$V = \frac{m}{\rho} = \frac{2}{0.5 \times 10^3}$$

$$V = 4 \times 10^{-3}$$

PART-2 : CHEMISTRY

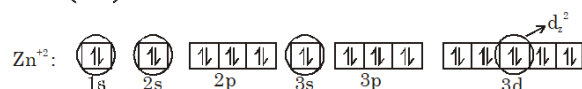
SECTION-I (i)

1. **Ans (B)**

$$\frac{n \times 56}{67200} \times 100 = 0.33$$

$$n \approx 4$$

2. **Ans (A)**



PART-2 : CHEMISTRY

SECTION-I (ii)

5. **Ans (A,D)**

$$\frac{(U_{\text{rms}})_{\text{O}_2, \text{TK}}}{(U_{\text{av}})_{\text{O}_2, \text{TK}}} = \frac{\sqrt{\frac{3RT}{M}}}{\sqrt{\frac{8RT}{\pi M}}} = \sqrt{\frac{3\pi}{8}}$$

6. **Ans (A,C)**

$$U_{\text{avg}} \propto \sqrt{\frac{T}{M}}$$

$$(U_{\text{avg}})_{\text{CH}_4} \propto \sqrt{\frac{400}{16}}; (U_{\text{avg}})_{\text{O}_2} \propto \frac{600}{32}$$

$$\text{KE} = \frac{3}{2}RT$$

PART-2 : CHEMISTRY

SECTION-II (ii)

7. **Ans (6)**

At node,

$$\Psi(r) = 0$$

$$\Rightarrow (1 - \sigma)(\sigma^2 - 8\sigma + 12) = 0$$

$$\sigma = 1, 2, 6$$

$$\text{Now } r = \frac{na_0}{2Z} \cdot \sigma$$

Hence, nodes will form at

$$r = \frac{na_0}{2Z}, \frac{na_0}{2Z} \times 2 \text{ and } \frac{na_0}{2Z} \times 6$$

(nearest)

(farthest)

$$= x$$

$$= y$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (A)**

D I L M R U

D 5!

I 5!

L 5!

M D 4!

M I 4!

M L 4!

M R D 3!

M R I D L U

M R I D U L

$$\text{Rank} = 5! \times 3 + 4! \times 3 + 3! + 2$$

2. **Ans (B)**

$$\begin{aligned} a^2 + b^2 + c^2 + ab + bc + ca &\leq 0 \\ \Rightarrow (a+b)^2 + (b+c)^2 + (c+a)^2 &\leq 0 \\ \Rightarrow a+b &= b+c = c+a = 0 \\ \Rightarrow a=b=c &= 0 \end{aligned}$$

3. **Ans (A)**

$$\begin{aligned} a &= 2\cos\theta \\ b &= 2\sin\theta \\ c &= \cos\alpha, d = \sin\alpha \\ ac + bd &= 2\cos(\theta - \alpha) \leq 2 \end{aligned}$$

4. **Ans (D)**

9 places 9 persons in 9! ways

PART-3 : MATHEMATICS

SECTION-I (ii)

5. **Ans (B)**

$$\begin{aligned} T_{r+1} &= {}^9C_r (2x)^{\frac{9-r}{2}} \left(\frac{3}{x^2}\right)^{r/3} \\ &= {}^9C_r 2^{\frac{9-r}{2}} 3^{r/3} \cdot x^{\frac{27-7r}{6}} \\ \Rightarrow r &= 3 \Rightarrow \text{coefficient} = {}^9C_3 \cdot 2^3 \cdot 3 \\ &= \frac{9 \times 8 \times 7}{6} \times 8 \times 3 = 84 \times 24 = 2016 \end{aligned}$$

6. **Ans (A,B,C,D)**

Any point P on the hyperbola $x^2 - y^2 = 4$ will be $(2\sec\theta, \tan\theta)$.

Equation of the normal at this point will be

$$2x\cos\theta + 2y\cot\theta = 8$$

$$x\cos\theta + y\cot\theta = 4$$

Its intersection point with transverse and conjugate axis will be $A(4\sec\theta, 0)$ and $B(0, 4\tan\theta)$ respectively.

centre is $(0, 0)$

$$\text{Now } PA = \sqrt{4\sec^2\theta + 4\tan^2\theta}$$

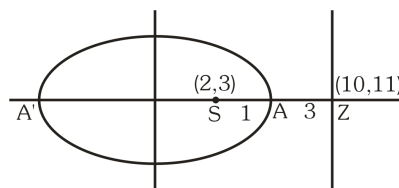
$$PB = \sqrt{4\sec^2\theta + 4\tan^2\theta}$$

$$PC = \sqrt{4\sec^2\theta + 4\tan^2\theta}$$

$$AB = \sqrt{16\sec^2\theta + 16\tan^2\theta} = 2\sqrt{4\sec^2\theta + 4\tan^2\theta}$$

$$\therefore PA = PB = PC = \frac{AB}{2}$$

7. **Ans (A,B,C)**



$$A \equiv \left(\frac{10+6}{4}, \frac{11+9}{4}\right) = (4, 5)$$

$$A' \equiv \left(\frac{10-6}{1-3}, \frac{11-9}{1-3}\right) = (-2, -1)$$

$$\text{Major axis, } AA' = \sqrt{36+36} = 6\sqrt{2}$$

$$\text{Minor axis } 2b = 2a\sqrt{1-e^2} = 2(3\sqrt{2})\sqrt{1-\frac{1}{9}} = 8$$

$$\text{Latus rectum} = \frac{2b^2}{a} = \frac{2 \cdot 16}{6\sqrt{2}} = \frac{8\sqrt{2}}{3}$$

8. **Ans (C,D)**

$$t_1 t_2 = -4$$

$$t_2 = -t_1 - \frac{2}{t_1}$$

$$t_1 t_2 = -t_1^2 - 2$$

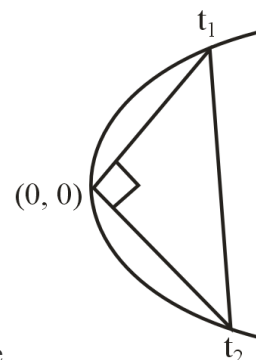
$$-4 = -t_1^2 - 2$$

$$t_1^2 = 2$$

$$t_1 = \sqrt{2}, -\sqrt{2}$$

$$\Rightarrow P(at_1^2, 2at_1) \text{ can be}$$

$$(4, 4\sqrt{2}) \text{ or } (4, -4\sqrt{2})$$



9. **Ans (A,B,C)**

$$N = 16200 = 2^3 \cdot 3^4 \cdot 5^2$$

$$(B) \text{ Divisors multiple of } 10 = (3)(4+1)(2) = 30$$

$$(C) \text{ Total - divisible by } 15$$

$$= (3+1)(4+1)(2+1) - (3+1)(4)(2)$$

$$= 60 - 32 = 28$$

$$(D) \text{ Even divisors} = (3)(4+1)(2+1) = 45$$

10. **Ans (A,B,C,D)**

$$P(3, 7), AB = 4$$

$$\Rightarrow AC = 4\sqrt{2}$$

Let equation of AC

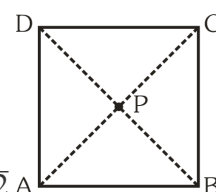
$$\frac{x-3}{\cos 45^\circ} = \frac{y-7}{\sin 45^\circ} = \pm 2\sqrt{2} A$$

$$(x, y) = (3 \pm 2, 7 \pm 2) \equiv (5, 9), (1, 5)$$

Similarly equation of BD

$$\frac{x-3}{\cos 135^\circ} = \frac{y-7}{\sin 135^\circ} = \pm 2\sqrt{2}$$

$$(x, y) = (3 \mp 2, 7 \pm 2) \equiv (1, 9), (5, 5)$$



PART-3 : MATHEMATICS

SECTION-II (i)

1. **Ans (10.00)**

Focus lie on x-axis, i.e., $y = 0$.

Solving with $2x + y = 12$, we get $x = 6$

Hence focus is $(6, 0)$

There is only one tangent at a distance of 1 unit from focus, which is possible only when this tangent is drawn at corresponding vertex.

Hence corresponding vertex will be $(5, 0)$

$$\Rightarrow a = 5$$

$$\text{Also, } ae - a = 1 \Rightarrow e = \frac{6}{5}$$

2. **Ans (1.20)**

Focus lie on x-axis, i.e., $y = 0$.

Solving with $2x + y = 12$, we get $x = 6$

Hence focus is $(6, 0)$

There is only one tangent at a distance of 1 unit from focus, which is possible only when this tangent is drawn at corresponding vertex.

Hence corresponding vertex will be $(5, 0)$

$$\Rightarrow a = 5$$

$$\text{Also, } ae - a = 1 \Rightarrow e = \frac{6}{5}$$

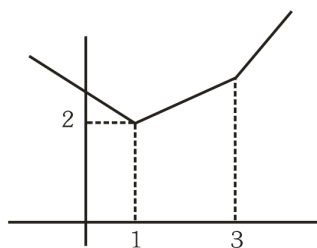
3. **Ans (32.00)**

$$\frac{2a}{e} = \frac{(2a)^2}{2ae} = \frac{16 \times 16}{8} = 32$$

4. **Ans (8.00)**

$$a = 8$$

5. **Ans (2.00)**



$$f(x) = |x-3| + 2|x-1|$$

$$= \begin{cases} -3x + 5 & x < 1 \\ x + 1 & 1 \leq x \leq 3 \\ 3x - 5 & x > 3 \end{cases}$$

$$f(x)|_{\min} = 2$$

6. **Ans (0.00)**

$$g(x) = -2x^2 + 3x - 1$$

$$g(x)|_{\max} = -\frac{D}{4a} = -\frac{9 - 4(-2)(-1)}{4(-2)} = \frac{1}{8}$$

$f(x)$ & $g(x)$ do not intersect at any point

$\therefore$ Equation $f(x) = g(x)$ does not have any solution.

PART-3 : MATHEMATICS

SECTION-II (ii)

7. **Ans (7)**

$$\text{Radical axis} \equiv S - S' = 0$$

$$\Rightarrow 2x(1-k) + 2y(k-0) + (5-k) = 0$$

is normal to given circle hence centre lies on radical axis

$$\Rightarrow 2(1-k) + 2(k-0) + 5-k = 0$$

$$2 - 2k + 2k + 5 - k = 0$$

$$\Rightarrow k = 7$$

8. **Ans (1)**

$$T_n = \frac{2n+1}{n^2(n+1)^2} = \frac{(n+1)^2 - n^2}{n^2(n+1)^2}$$

$$T_n = \frac{1}{n^2} - \frac{1}{(n+1)^2}$$

$$T_1 + T_2 + T_3 \dots + T_n = 1 - \frac{1}{(n+1)^2}$$

then sum of ∞ terms = 1

9. **Ans (3)**

$$\cos^4 x = 1 + \sin^3 x$$

$$\cos^2 x = 1$$

$$x = n\pi$$

solution are $0, \pi, 2\pi$



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REVIEW TEST

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JEE(Main + Advanced) : NURTURE COURSE [PHASE : S(TEPS)]

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,B,D	A,C,D	A,C	B,C	B,C,D	A,B,C
SECTION-I (ii)	Q.	7	8	9	10		
	A.	A	A	A	C		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	10.00	4.00	2520.00	1150.00	1.66 to 1.67	0.20
SECTION-II (ii)	Q.	7	8	9			
	A.	9	40	6			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,C,D	A,C	A,C	A,C	A,B,C	A,B,C,D
SECTION-I (ii)	Q.	7	8	9	10		
	A.	D	D	D	C		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	3.00	5.00	31.60	2.50	2.20	0.00
SECTION-II (ii)	Q.	7	8	9			
	A.	4	5	5			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,B,C,D	B,D	A,B,C	A,B	B,C	A,C
SECTION-I (ii)	Q.	7	8	9	10		
	A.	D	A	C	D		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	3.00	191.00	756.00	216.00	737.00	0.25
SECTION-II (ii)	Q.	7	8	9			
	A.	26	2	6			

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,B,D)

$$\Delta U_A = \Delta U_B$$

$$W_A > W_B \text{ (area under curve)}$$

$$\therefore Q_A > Q_B$$

3.

Ans (A,C)

Work done by gas from D to A

$$= nRT \ln \left(\frac{2v_0}{v_0} \right) = RT_0 \ln 2$$

Work done by gas from B to C

$$= R(2T_0) \ln \left(\frac{v_0}{2v_0} \right)$$

$$= -2RT \ln(2)$$

$$W_{\text{Net}} = -RT \ln(2)$$

In cyclic process $\Delta u = 0$

$$\text{So } \Delta Q = W$$

$$\Delta Q = -ve$$

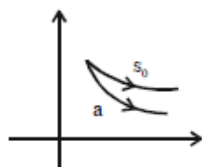
4. **Ans (B,C)**

Process A-B is isobaric while process BC is isothermal.

5. **Ans (B,C,D)**

$$\vec{L} = \vec{r} \times m\vec{v}$$

6. **Ans (A,B,C)**



PART-1 : PHYSICS

SECTION-I (ii)

7. **Ans (A)**



$$R = \frac{\ell}{kA}$$

$$\frac{R_A}{R_B} = \frac{\ell_A}{\ell_B} \frac{k_B}{k_A} = \frac{\ell \times k}{\ell 3k} = \frac{1}{3}$$

8. **Ans (A)**

Since both resistance are in series

$\Rightarrow i \rightarrow$ same

$V = iR$
potential drop

$\Delta T = iR_{\text{thermal}}$
temperature drop

$$\frac{\Delta T_1}{\Delta T_2} = \frac{R_1}{R_2} = \frac{1}{3}$$

9. **Ans (A)**

$$U = -\frac{x^4}{4} - \frac{x^3}{3} + x^2$$

$$F = -\frac{dU}{dx} = x^3 + x^2 - 2x$$

$$= x(x^2 + x - 2)$$

$$= x(x-1)(x+2)$$

For equilibrium $F = 0$

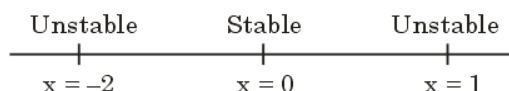
$$x = -2, x = 0, x = 1$$

$$\frac{dU}{dx} = -x^3 - x^2 + 2x \quad \frac{d^2U}{dx^2} = -3x^2 - 6x + 2$$

$$\text{At } x = 0, \quad \frac{d^2U}{dx^2} = 0 - 0 + 2 > 0 \text{ (stable)}$$

$$x = 1, \quad \frac{d^2U}{dx^2} = -3 - 6 + 2 = -7 < 0 \text{ (Unstable)}$$

$$x = -2, \quad \frac{d^2U}{dx^2} = -12 + 12 + 2 = 2 > 0 \text{ (Stable)}$$



10. **Ans (C)**

$$U = \frac{-x^4}{4} - \frac{x^3}{3} + x^2$$

at $x = 0, U = 0 \Rightarrow T.E = K.E. + P.E.$

$$16 = K.E. = \frac{1}{2}mv^2$$

$$v = 4 \text{ m/s}$$

PART-1 : PHYSICS

SECTION-II (i)

3. **Ans (2520.00)**

$$\lambda_m T = \text{constant}$$

$$\text{Emissive power} \propto \sigma T^4 \propto \frac{\sigma}{\lambda_m^4}$$

6. **Ans (0.20)**

at $t = 10 \text{ sec}$

$$\text{Area} = \frac{1}{2} \times 2 \times 10 = v_f - v_0$$

$$= t = 10, v = 10 \text{ m/s}$$

$$k = \frac{1}{2}mv^2 = \frac{1}{2} \times 10 \times (10)^2$$

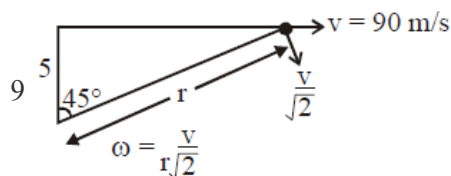
$$P = mv = 10 \times 10 = 100$$

$$\frac{P}{k} = \frac{100}{500} = 0.20$$

PART-1 : PHYSICS

SECTION-II (ii)

7. **Ans (9)**

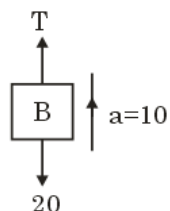


8. Ans (40)

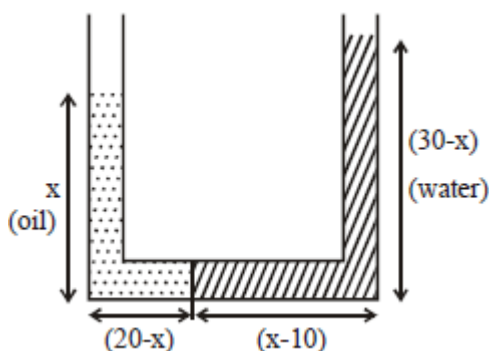
$$a = \frac{100 - 20 - 20 - 10}{5} = 10$$

$$T - 20 = 2 \times 10$$

$$T = 40$$



9. Ans (6)



$$\therefore \rho_{\text{oil}} g x + \rho_{\text{oil}} (20 - x) a + \rho_{\text{water}} (x - 10) a = \rho_{\text{water}} g (30 - x)$$

$$\therefore x = 12$$

$$\therefore \Delta H = (30 - x) - x = (30 - 2x) = 6 \text{ cm}$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (A,C,D)

For exothermic reaction,

$T \uparrow$, $K_p \downarrow$, reaction goes backwards

on decreasing the pressure, reaction moves towards more no. of moles.

2. Ans (A,C)

Theory based

4. Ans (A,C)

Ans. (A,C)

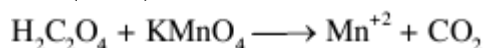
PART-2 : CHEMISTRY

SECTION-II (i)

3. Ans (31.60)

$$1 \times 1 = \frac{w}{(158/5)} \Rightarrow w = 31.6 \text{ gm}$$

4. Ans (2.50)



eq. of $\text{H}_2\text{C}_2\text{O}_4$ = eq. of KMnO_4

$$n \times 2 = 1 \times 5$$

$$n = \frac{5}{2}$$

PART-2 : CHEMISTRY

SECTION-II (ii)

7. Ans (4)

Eq. NaOH = Eq. H_2SO_4 + Eq. HCl

$$n \times 1 = NV_2 + NV_2$$

$$n \times 1 = 0.1 \times 2 \times 0.1 + 0.2 \times 0.4 = 0.1$$

$$n_{\text{NaOH}} = 0.1$$

wt. of NaOH = 4g.

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,B,C,D)

Use $D \geq 0$

$$16 + 4\log_2 a \geq 0$$

$$\log_2 a \geq -4 \Rightarrow a \geq \frac{1}{16}$$

2. Ans (B,D)

Let the slope of the required line be m

$$\text{So, } \left| \frac{m + \sqrt{3}}{1 - \sqrt{3}m} \right| = \sqrt{3} \Rightarrow m = 0, \sqrt{3}$$

$$\text{Hence, } y + 2 = 0 \text{ or } y + 2 = \sqrt{3}(x - 3)$$

3. Ans (A,B,C)

$$\left(x^7 - \frac{3}{x} \right)^n \therefore T_{r+1} = {}^nC_r (-3)^r (x)^{7n-8r}$$

$$r = \frac{7n}{8} \therefore n \text{ is multiple of } 8$$

$$\therefore n \text{ can be } 8, 16, 24$$

4. Ans (A,B)

A,B

5. Ans (B,C)

$$(B) {}^8C_4 \cdot 4!$$

$$(C) {}^7C_4 \cdot 4!$$

6. Ans (A,C)

$$S_n = 2n^2 + 3n$$

$$S_{n-1} = 2(n-1)^2 + 3(n-1)$$

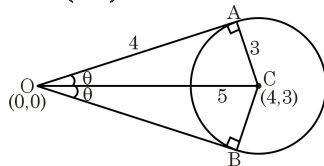
$$a_n = S_n - S_{n-1} = 4n + 1$$

$$\therefore a_1 = 5, a_2 = 9 \text{ and } d = 4$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. **Ans (D)**



$$\alpha = 2\theta$$

$$\sin \alpha = \frac{2 \tan \theta}{1 + \tan^2 \theta} = \frac{2 \cdot \frac{3}{4}}{1 + \frac{9}{16}} = \frac{24}{25}$$

8. **Ans (A)**

Circle with OC as diameter.

9. **Ans (C)**

$$\text{Area of ellipse} = \pi ab = \pi \sqrt{32.18} = 24\pi$$

10. **Ans (D)**

Normal : (Given)

$$4x - 3y = 7 \quad \dots (1)$$

$$\text{Ellipse : } \frac{x^2}{32} + \frac{y^2}{18} = 1$$

let point P : (x_1, y_1)

$$\therefore \text{Normal : } \frac{a^2 x}{x_1} - \frac{b^2 y}{y_1} = a^2 - b^2$$

$$\Rightarrow \frac{32x}{x_1} - \frac{18y}{y_1} = 14$$

$$\Rightarrow (32y_1)x - (18x_1)y = 14x_1y_1 \quad \dots (2)$$

compare (1) & (2)

$$\frac{32y_1}{4} = \frac{-18x_1}{-3} = \frac{14x_1y_1}{7}$$

$$\Rightarrow 8y_1 = 6x_1 = 2x_1y_1$$

$$\Rightarrow (x_1, y_1) \equiv (4, 3) \text{ Ans.}$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. **Ans (3.00)**

EEE, HHHHH, S_1, S_2, S_3

1E & 1H & 1 science book

$$1 \times 1 \times {}^3C_1 = 3$$

2. **Ans (191.00)**

EEE, HHHHH, S_1, S_2, S_3

$$4 \times 6 \times 2^3 - 1$$

3. **Ans (756.00)**

Required no of ways

$$\lambda = \frac{|6|}{|2|} \cdot {}^7C_3 |3| = 75600$$

$$\Rightarrow \frac{\lambda}{100} = 756$$

4. **Ans (216.00)**

Required ways

$$k = \left(\frac{|6|}{|2|} \cdot |3| \right)$$

$$k = 2160 \Rightarrow \frac{k}{10} = 216$$

5. **Ans (737.00)**

$$\text{Sum} = 65 + 62 + 59 + 56 + \dots + 2$$

$$= \frac{22}{2} (67) = 11 \times 67 = 737$$

$$S_n = \frac{n}{2} (130 + (n-1)(-3))$$

6. **Ans (0.25)**

First negative term, $k = -1$

$$\therefore \left| \frac{k}{4} \right| = \left| \frac{-1}{4} \right| = 0.25$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. **Ans (26)**

$$T_{r+1} = {}^{100}C_r 5^{\frac{100-r}{2}} (11)^{r/4}$$

for rational terms set of values of r is $\{0, 4, 8, 12, \dots, 100\}$

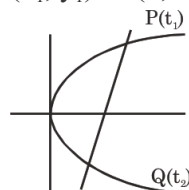
$$\therefore n(\{0, 4, 8, 12, \dots, 100\}) = 26$$

8. **Ans (2)**

$$T = S_1$$

$$yy_1 - 2a(x + x_1) = y_1^2 - x_1$$

$$(x_1, y_1) \Rightarrow (2, 1)$$



$$y - \frac{2}{4}(x + 2) = 1 - 2$$

$$4y - 2x = 0$$

$$x = 2y \Rightarrow \text{solve with parabola}$$

$$y^2 = 2y$$

$$\text{so, } y = 0, \quad y = 2$$

$$\& x = 0, \quad x = 4$$

$$P(0, 0) \quad Q(4, 2)$$

$$PQ = \sqrt{4 + 16} = 2\sqrt{5}$$

9. **Ans (6)**

"KOLAVARIDI"

$$n_1 = \frac{10!}{2! 2!}, \quad n_2 = 6! \times \frac{5!}{2! 2!}$$

$$\frac{n_1}{n_2} = 42 \Rightarrow \text{sum of digits} = 6$$